

ENGG3112 : STAGE-1

SUSTAINABLE TEACHING LABS

1. Background Research on the Project Client:

The **Faculty of Engineering** at the **University of Sydney** is one of leading institutions in Australia. Consisting of seven schools, it collaborates together to drive innovation in engineering-related fields. Their vision is to create world-class infrastructure for research and education, while supporting collaborative decarbonisation innovative programs in Australia. They are committed to achieving net-zero emissions by 2030, as part of the University's initiative namely, **Sustainability Strategy**. They are dedicated to enhancing the student experience, for both - incoming and ongoing students. The faculty aims to expand access to practical, hands-on training in skills relevant to their discipline. They are also mindful of minimising emissions of the operations as they expand the laboratories and workshop spaces. Additionally, they strive to create a more inclusive environment for all the students irrespective of their cultural backgrounds, gender and physical abilities. Moving forward, they are determined to reduce the environmental impact while maintaining the global standards in teaching and research. As a shareholder, all their views and priorities coincide with the project's focus. The key shareholders of this project though are: Dr Shuying Wu and Professor Matthew Cleary belonging to the School of Aeronautical, Mechanical and Mechatronics Engineering (AMME) who want to expand the current capacity of first year undergraduate workshop experience delivered in J07 and second year dynamic laboratories for AMME2500 in J07, S173 Halliday Lab / Maker Space, respectively. The key stakeholders also seem to be benefitted from the expanded teaching and learning spaces for their specific courses. [1][2][3][4]

2.1. Self-Interpretation of Project Brief:

The project's main focus appears to be expanding first-year undergraduate workshop experiences in J07 and second-year dynamics laboratories for AMME2500 in J07, S173 Halliday Lab/Maker Space, managed by the School of AMME. The outdated infrastructure imposes numerous inconveniences to the staff as well as the students. With enrollment steadily increasing over the past decade or so, it's essential to upgrade these learning and teaching spaces to provide a world-class learning and fun experience. Upgrading a lab infrastructure is already a tedious process, especially with a limited budget. Frequent breakdowns and outdated systems or equipments hampers one's experience and hence renewal is needed from time-to-time for long-term productivity and extended reliability. Now that people are more aware and active about reducing their carbon footprints, it is necessary for the department to follow the same suit.

Laboratory operations significantly contribute to carbon emissions, hindering the University's plan to achieve a net-zero target by 2030, therefore, substantial upgrades are needed. Specialised foundations have to be made to tackle this hazard waste problem which endangers our environment, safety albeit following the regulatory guidelines at the same time. Considering that there are people of varying physical abilities in a course, we should keep the systems upgraded so as to avoid any possible hazardous incident and to provide hassle-free participation. Outdated equipments could even sometimes lead to non-compliance of safety standards hence, it is significant that we don't compromise the safety of others and instead upgrade the infrastructures. Modern infrastructure provides flexibility in interactive learning to the student community and promotes collaboration and communication between them and the staff, while also maintaining the global learning and teaching standards. [1][2][3][5]

2.2. Problem Definition Statement:

The Faculty of Engineering's goal of establishing sustainable teaching labs and workshop spaces is commendable. However, several significant challenges must be overcome before the project can be successfully brought to completion. The primary obstacle is the expansion of functional capacity within the limited space of existing laboratories and workshops. Furthermore, it is crucial to monitor and track the environmental impact of these areas, including energy consumption, waste production, and consumables. Identifying the opportunities for reducing environmental footprints is also essential. Lastly, improving the accessibility of these labs and spaces for neurodiverse students with diverse physical abilities is another key obstacle that requires attention, ensuring that these areas are inclusive for all hence enhancing the quality of practical education. All these problems need to be solved in order to address today's needs. Although new innovative and sustainable solutions are required to provide a holistic environment for upcoming students while also completing the future needs.

3. Precedents:

There are precedents of the current problem in many other organisations outside the University of Sydney.

The first can be seen in **Harvard University**, lab sustainability aims to foster health safety, resource efficiency and waste reduction strategies to minimise environmental footprint while promoting scientific advancements. They conducted a program named 'Shut the Sash' which was launched in 2005 when the Department of Chemistry and Chemical Biology began exploring new ways to reduce waste productions. It promotes keeping the fume hoods closed when not in use and the hood should never remain open for more than 5 hours. Each of the labs has a customised CFM goal based on number of fume hoods (with HVAC system), number of researchers, and the type of research conducted. A lab that achieves their goal goes into a lottery for a party, this acts as an incentive which promotes the students to together reach for their

common goal of reducing environmental footprints. Year after year, the department saw that this program saved Harvard energy and cost savings of \$200,000 per year with greenhouse gas emissions savings at 300+ metric tonnes of carbon dioxide equivalent. They also collaborate with many non-profit organisations and vendors to pilot innovative technologies aimed at decarbonising the built environment and reducing campus waste. The Engineering department has also made bioretention basins and swales to capture rainwater, rooftop gardens are also being made in order to reduce stormwater runoff, cool roof surfaces and increase green spaces on campus promoting greener outdoor spaces to students. Free Fixit Clinics are also built on campus for people to bring their broken items to the “fixers” who can help diagnose and hopefully fix the items, hence contributing to the culture of reuse. Their plan to replace the shuttle-bus fleet to electric and improve sustainable transportation opportunities like installation of on-campus EV charging stations, programs that provide benefits if using bicycle, walking and carpooling. The Harvard Living Lab: Sustainability Accelerator brings the students and staff together to test and validate innovative new solutions and ideas across all disciplines hence promoting coordination and communication and bridging the gap between Harvard employees and its students. [6][7][8][9][10][11]

The **University of Technology, Sydney (UTS)** has introduced an underground Hive Superlab with a focus on collaboration, capable of accommodating up to 270 students. Each coloured space in the lab is a distinct zone equipped with specialist AV technology, including bone-conduction headphones that allow lab-users to hear their facilitator while engaging in conversations with their peers. With the ability to run up to seven classes simultaneously, students from different disciplines can learn side by side, promoting the opportunity for cross-pollination of knowledge and ideas. This inclusive learning environment also offers hands-on and practical experiences, making it accessible to students from various degrees and physical abilities, thereby enhancing participation and communication. In addition to the Hive Superlab, UTS also has a Science SuperLab that boasts modern equipment and technology. This lab can accommodate over 200 students from 12 different classes, enabling practical learning across multiple disciplines. With approximately 60 breakout rooms, some of which are equipped with smart boards, students have plenty of resources at their disposal. Both labs also offer 24/7 access, ensuring accessibility for all students. One of UTS's impressive facilities is the Data Arena, a 360-degree interactive data visualisation space. This innovative facility allows for the visualisation of complex information, trend identification, problem-solving, and is user-friendly. Through 3D animation, the learning experience becomes engaging and stimulating, fostering innovation and creativity. By visualising complex data in front of you, you can gain a comprehensive understanding of the problem at hand. UTS has truly demonstrated that these innovative and modern spaces can elevate your learning journey to new heights, making it truly amazing. [12][13][14][15][16]

4. References:

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